Data Analysis with Python

Python 3

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Data Analysis with PythonIntroductionThis notebook demonstrates basic data analysis techniques using Python's pandas library. We will explore a dataset, perform statistical analysis, and create visualizations.Learning Objectives• Load and explore datasets
• Perform descriptive statistics
• Create data visualizations
• Draw insights from data

1. Setting Up the EnvironmentFirst, we import the necessary libraries for our analysis.

Code:

```
1 | import pandas as pd 2 | import numpy as np 3 | import matplotlib.pyplot as plt 4 | 5 | # Configure display options 6 | pd.set_option('display.max_columns', 10) 7 | pd.set_option('display.width', 1000)
```

2. Creating Sample DataFor this demonstration, we'll create a sample dataset representing student performance.

Code:

```
1 | # Create sample student data 2 | data = { 3 | 'Student ID': [1, 2, 3, 4, 5], 4 | 'Math Score': [85, 78, 92, 70, 88], 5 | 'Reading Score': [90, 82, 95, 75, 89], 6 | 'Study Hours': [5, 3, 7, 2, 6] 7 | } 8 | 9 | df = pd.DataFrame(data) 10 | df
```

3. Descriptive StatisticsLet's examine the statistical summary of our dataset.

Code:

```
1 | # Display descriptive statistics 2 | df.describe()
```

4. Data Analysis FunctionsWe can create reusable functions for common analysis tasks.

Code:

```
1 | def calculate_correlation(df, col1, col2): 2 | """Calculate Pearson correlation between two columns.""" 3 | return df[col1].corr(df[col2]) 4 | 5 | def get_performance_category(score): 6 | """Categorize performance based on score.""" 7 | if score >= 90: 8 | return 'Excellent' 9 | elif score >= 80: 10 | return 'Good' 11 | elif score >= 70: 12 | return 'Average' 13 | else: 14 | return 'Needs Improvement'
```

Code:

```
1 | # Analyze correlations 2 | math_reading_corr = calculate_correlation(df, 'Math Score', 'Reading Score') 3 | hours_math_corr = calculate_correlation(df, 'Study Hours', 'Math Score') 4 | 5 | print(f"Correlation between Math and Reading scores: {math_reading_corr:.3f}") 6 | print(f"Correlation between Study Hours and Math Score: {hours_math_corr:.3f}")
```

5. Key FindingsSummaryBased on our analysis:1. **Strong Positive Correlation**: Math and Reading scores are highly correlated (r = 0.985)
2. **Study Impact**: Study hours show strong positive correlation with performance

3. **Average Performance**: Mean math score is 82.6, reading score is 86.2Recommendations• Encourage increased study time for improved performance
• Consider integrated math-reading curriculum
• Provide additional support for students scoring below 75

Code:

```
1 | # Final summary 2 | print("\n=== Analysis Complete ===") 3 | print(f"Dataset: Student Performance") 4 | print(f"Total Records: {len(df)}") 5 | print(f"Analysis Date: 2026-03-18")
```